

Instructions

There are **25 total points**. When asked to provide your answer within a figure or table, be careful to not exceed box boundaries. Bubbles must be filled out completely: ● is correct, ☑ ● ✕ are incorrect All answers must be given within the provided circles, answer boxes, figures or tables.

1. [1 point]: Write your full name in the box to acknowledge the instructions.

Use Cases

2. [4 points]: Machine learning has become a useful tool in modeling and predicting the performance of networked systems. In class and in the assignments, you studied how machine learning can be used to predict various aspects of application quality. For the case of *video on demand*, give an example of an (1) **three** application quality metrics that machine learning is often used to predict; (2) **one** feature that is often used to predict these metrics.
- (Answer inside the box)

3. [4 points]: Internet scanning is a technique to discover vulnerabilities of machines that are connected to the Internet. One form of scanning that we studied was HTTP (Web) scanning, as in the Log4j vulnerability scan. As we studied, the HTTP traffic generated by the scan looked “different” than regular Web traffic. Give two features that could be used to distinguish between HTTP scanning traffic and regular Web traffic.
- (Answer inside the box)

Data Acquisition

4. [2 points]: With access to unencrypted DNS queries and responses and knowledge about which DNS names correspond to a particular service (e.g., Netflix), it is possible to extract IP addresses for that service.

☐ Yes ☐ No

Alyssa P. Hacker wants to extract all Netflix traffic from packet traces, but reasoning that the code for matching DNS names to IP addresses is cumbersome, she decides instead to simply hard-code the IP addresses for a particular service. Alyssa had constructed her initial IP address list by looking up domain names for Netflix here in Chicago. When she deploys her code to networks in other locations around the world, she is surprised to find that her code is not extracting any Netflix traffic!

Initials: ____

5. [2 points]: Give one reason why Alyssa's code may not be capturing Netflix traffic in other locations.

(Answer inside the box)

6. [3 points]: List three advantages to passive Internet measurement over active Internet measurement.

(Answer inside the box)

Feature Engineering

7. [2 points]: Explain **one advantage** of using domain knowledge to engineer features for model input as opposed to using byte representations of traffic as input. (There are several correct answers.)

(Answer inside the box)

8. [5 points]: The University of Chicago ITS wants to identify compromised devices on its network and provides you with a labeled traffic trace of benign and compromised devices to build a model to detect them. Your model achieves 99% accuracy on the training data, and the most important feature is the set of domain names that each device contacts. Your model works well on the training set and for the first few days, but a week later, it is failing to detect many of the compromised devices. (1) Why might this feature have worked initially? (2) Why might it have stopped working?

(Answer inside the box)

Feedback

9. [1 point]: Interest (1=Boring!; 10=Amazing!):

Difficulty (1=Too easy; 10=Too hard):

10. [1 point]: 1. One thing you like. 2. One suggestion for improvement:

(Answer inside the box)

Initials: ____